

Human-Parity Grading of DESI-Resolution Strong-Lens Candidates: A Truth-Referenced Comparison of Expert Graders and Machine Systems

Greg Benson*

Xiaosheng Huang†

July 2026

The capstone of the parity program: a pre-registered, truth-anchored test of whether any machine system grades DESI-resolution lens candidates at trained-human level — what the human baseline turns out to be worth against truth, which machine systems survived a frozen evidence gate, and what today’s fourteen refutations can and cannot decide

Abstract

The companion human-baseline report made “as good as a human grader” a number for the DESI Legacy Surveys strong-lens search (Huang et al., 2020, 2021; Inchausti Reyes et al., 2025; Storfer et al., 2024): two trained graders agree at $\kappa_{\text{qw}} = 0.42$, an individual grader matches the published consensus at 0.776, an independent expert team at 0.29, and confirmation purity among decided follow-ups is high and flat in grade (95.2/92.0/90.9%). This report is the program’s capstone: it builds the apparatus to decide whether a machine meets those targets, and runs the comparison. The design is two-endpoint — E2 (primary, truth-referenced): machine score and human ordinal grade as *paired* predictors of confirmed-vs-refuted follow-up truth; E1 (secondary): agreement with the consensus — executed with a leak firewall and a frozen, score-once evidence gate. Three findings frame the result. *First*, the human grade itself is statistically consistent with chance against truth on the followed-up pool (AUC = 0.577 [0.396, 0.757]; $n = 130$ decided, only 8 refuted): the *selection* of candidates did the work, and the fine grade adds almost no incremental truth signal — so E2 “parity” is a low bar concealing a data problem, not a high one. *Second*, the data to decide small margins does not yet exist: auditing the Euclid Q1 truth chain (a withdrawn NISP-spectroscopy paper excluded; SLDE-A’s per-object verdicts unpublished) leaves exactly 14 per-object refutations in existence across all campaigns, a paired non-inferiority noise floor of ~ 0.24 in AUC today, falling to 0.078 only at reader-study scale. *Third*, the machines nonetheless moved: on the frozen 259-row gate a Qwen3.5-27B student fine-tuned directly on the human grade catalog reaches 0.644 [0.550, 0.733] on the hardest contrast (published A/B lenses vs. grade-D human rejects) — tying the frozen CNN’s 0.646 [0.557, 0.734] and reaching 0.817 [0.757, 0.873] on vetting, a +0.11/+0.24 training delta over its zero-shot control — where v4-style Claude distillation had failed; and giving a Claude grader the human graders’ own information set produced the program’s first statistically significant tier-1 lever ($\Delta\text{AUC} = +0.151$ [+0.020, +0.280] over the deployed grader), whose *absolute* level on the hard contrast then failed to replicate at the gate (0.487) — the disjoint-draw discipline doing exactly its job. The pre-registered Phase D comparison (paired stratified bootstrap ΔAUC against the human grade,

*benson@cs.usfca.edu; Department of Computer Science, University of San Francisco, CA 94117, USA

†Department of Physics & Astronomy, University of San Francisco; and Physics Division, Lawrence Berkeley National Laboratory

DeLong test, non-inferiority margin $\delta_0 = 0.05$; agreement κ_{qw} ; an Euclid-field context arm) then delivers the program’s answer: on the 130 truth-decided candidates, **both the student** (AUC = 0.685, $\Delta\text{AUC} = +0.108$ [−0.028, +0.241]) **and the frozen CNN** ($\Delta\text{AUC} = +0.182$ [−0.011, +0.425]) **are formally non-inferior to the expert grade** — with point estimates that *favor* the machines, though superiority is not established (DeLong $p \approx 0.13$) and only eight refuted negatives carry the comparison. The machines do not reproduce the letter grade itself ($\kappa_{\text{qw}} 0.044$), and the student’s truth signal does not transfer to the harder Euclid-selected case mix (0.509). The defensible deployment today is therefore score-based vetting through calibrated operating points — the letter grades stay human.

1 The question, made decidable

The DESI Legacy Surveys lens search publishes candidates as expert A/B/C grades on neural-network recommendations (Huang et al., 2020, 2021; Inchausti Reyes et al., 2025; Storfer et al., 2024); the prospect that an automated grader could stand in for the expert step — “as good as a human grader” — has hung over the program since LENSJUDGE Part I¹ measured a Claude-agent grader at $\kappa \approx 0$ against the consensus. That phrase is empty until two things exist: a measured human level, and a comparison design that could falsify the claim. The parity program (approved plan, July 2026) supplies both, in four phases. **Phase A** measured the human baseline from published data alone — condensed in §2, derived in full in the companion report.² **Phase B** built a truth-anchored bench and did the power arithmetic (§3). **Phase C** trained and froze the machine systems behind a leakage firewall and scored each *once* on a frozen gate (§4). **Phase D** is the pre-registered parity comparison itself (§5).

Two endpoints, fixed before any machine was scored, carry the claim. **E2 (primary, truth-referenced)**: on candidates whose follow-up is decided, compare the machine score and the human ordinal grade as *paired* predictors of confirmed-vs-refuted truth, with a non-inferiority margin. Truth — spectroscopy and space-resolution imaging — is the referee, not the humans being compared against. **E1 (secondary, agreement)**: κ_{qw} of machine grades against the published consensus, read against the measured human anchors. E2 is primary by design: agreement with a consensus that §3.3 shows is itself nearly uninformative about truth on this pool would be a strange thing to optimize.

Discipline mattered as much as design. Every training pool was firewalled against the bench by name and position (§4.1); model and operating-point selection happened only on a selection split (**valsel**); the frozen gate was touched exactly once per frozen system; and Phase D’s predictors and analysis code were fixed before the bench was scored. The one selection-vs-gate non-replication the program produced (§4.5) is reported as prominently as the wins, because it is the discipline visibly working.

2 The human baseline, in brief

The companion report² recovered the baseline from published data alone; Table 1 collects the numbers this report scores machines against. *Reliability*: the two Huang et al. (2021) graders gave the same 1–4 score on only 55.4% of accepted candidates, a mutual $\kappa_{\text{qw}} = 0.42$ [0.37, 0.47]; an individual grader matches the published consensus at $\kappa_{\text{qw}} = 0.776$ [0.757, 0.793] (an upper bound —

¹LENSJUDGE: *An Agentic Strong-Lens Candidate Grader*, companion report, <https://usfcs-edu.github.io/agentic-lensing/current/lensjudge/>.

²*The Human Baseline for Strong-Lens Grading*, companion report, <https://usfcs-edu.github.io/agentic-lensing/current/lensjudge-human-baseline/>

	anchor	value
E1 (agreement)	individual grader vs. consensus (upper bound)	$\kappa_{\text{qw}} = 0.776$ [0.757, 0.793]
	the two graders’ mutual agreement	$\kappa_{\text{qw}} = 0.420$ [0.371, 0.468]
	independent expert team (SuGOHI, $n = 103$)	$\kappa_{\text{qw}} = 0.288$ [0.121, 0.444]
E2 (truth)	decided purity by grade (A/B/C)	95.2 / 92.0 / 90.9%
	grade-A refutations (VLT/MUSE)	4 of 83 decided
	human grade as truth predictor (§3.3)	AUC = 0.577 [0.396, 0.757]

Table 1: The measured human baseline (Phase A, companion report²) plus the Phase B truth-prediction headline this report adds. All reliability numbers are upper bounds (the published catalog is truncated at acceptance).

each grader is half the consensus); independent expert teams grading the same candidates land at $\kappa_{\text{qw}} = 0.288$ [0.121, 0.444] (SuGOHI committee, $n = 103$; Sonnenfeld et al. 2018) and 0.165 (Euclid Q1 panel, $n = 24$; Euclid Collaboration et al. 2025b) — consistent with the controlled-experiment literature (Cañameras et al., 2021; Rojas et al., 2023). *Validity:* across every published per-target follow-up campaign (Agarwal et al., 2025; Barone et al., 2025; Huang et al., 2026; Huang et al., 2025; Lin et al., 2025; Shu et al., 2025; Sonnenfeld et al., 2018; Tran et al., 2022), 162 graded candidates have follow-up and 130 are decided; confirmation purity is $79/83 = 95.2\%$ (A), $23/25 = 92.0\%$ (B), $20/22 = 90.9\%$ (C) — high everywhere, flat in grade, and pointedly not 100% for grade A: four grade-A candidates are spectroscopically refuted by VLT/MUSE (Lin et al., 2025), every one an arc-shaped feature at a disqualifying redshift — the morphology-only failure mode that grades assigned from $1.3''$ -seeing pixels cannot see.

3 The truth-anchored parity bench

3.1 Two arms

The bench (`eval/build_parity_bench.py`, seed 2026, SHA-pinned CSVs) has two arms, matching the two things a parity claim needs: a *paired* human-vs-machine comparison, and behavior at survey scale.

Arm 1 (paired, DESI resolution) is every truth-followed graded candidate: the 162 candidates with published follow-up (99 A, 30 B, 33 C), of which 130 are decided — 122 confirmed lenses and 8 spectroscopic refutations. To these we add 750 non-truth augmentation rows (250 grade-D human rejects, 500 random galaxies) so that operating points and score distributions are estimated in a realistic candidate mix; augmentation rows never enter a truth endpoint. Total: 912 rows.

Arm 2 (machine at scale, Euclid field) is all 2,584 Euclid Q1 Strong Lensing Discovery Engine (SLDE-A) catalog positions (Euclid Collaboration et al., 2025b) served as DESI Legacy Surveys DR10 cutouts (Dey et al., 2019) — the machine grades ground-based pixels of candidates an independent ~ 10 -expert panel graded at $0.1''$ (309 A, 267 B, 2,008 C, with the panel’s `expert_score` and vote counts carried) — plus 21 decided SLDE-B objects outside the Q1 catalog, for 2,605 rows. Of these, 44 carry per-object truth (38 lens, 6 refuted; modeling tier — see §3.2). Cutout staging is complete on both arms (162/162; 2,605/2,605).

3.2 Euclid truth integrity: two findings that shrank the truth set

The plan projected a few hundred Euclid-side adjudications. Auditing the actual truth chain reduced that to 44, for two reasons we record because they change what any group can currently claim.

(1) The “NISP spectroscopy of Q1 lenses” paper is withdrawn. The single-author paper offering deflector redshifts for Q1 candidates³ cannot anchor a truth endpoint. (2) **SLDE-A’s per-object verdicts are unpublished.** The discovery-engine paper reports only aggregates (374 modeled, 315 judged lens, 59 judged non-lens), and the Zenodo release’s `unsuccess.zip` mixes modeling failures, non-lenses, and ambiguous cases with no verdict field (Euclid Collaboration, 2025; Euclid Collaboration et al., 2025b). Only SLDE-B (Euclid Collaboration et al., 2025a) publishes per-object verdicts: 38 systems confirmed by automated modeling plus expert inspection, 6 ruled out, 11 model-failed (inconclusive). The recoverable per-object Euclid truth today is therefore 44 decided systems, not ~ 390 — and the program-wide ledger of per-object *refutations*, the currency of any truth-referenced comparison, stands at 8 (arm-1 spectroscopic) + 6 (SLDE-B modeling) = 14, anywhere in the literature.

3.3 Power: what today’s truth can decide

Before scoring any machine we asked what the bench can decide (`parity/power_check.py`). The headline reframes the whole program: **the human ordinal grade, scored against confirmed-vs-refuted truth on the 130 decided candidates, has AUC = 0.577 [0.396, 0.757] — statistically consistent with chance.** High purity in every grade plus flatness across grades means the *selection* of candidates for publication and follow-up did the discriminative work; conditional on being followed up, the fine grade adds almost no incremental truth signal. (Caveats: curated, best-first follow-up compresses the score range; the grade has three levels; there are only 8 negatives.)

The same arithmetic bounds any verdict. Under an equal-AUC binormal model with paired correlation $\rho = 0.6$, today’s 122/8 split gives a paired- Δ AUC standard error of 0.095: the smallest decidable non-inferiority margin is ~ 0.24 — coarse screening only. The ladder to a real verdict: ~ 30 refutations (the scale the grade-agnostic DESI Secondary Target Program, 2,157 systems, should deliver first; Huang et al. 2025) brings it to 0.13; an Euclid-era 222/60 set to 0.10; reader-study scale (250/100) to 0.078. **Estimation yes, small-margin verdict no** — and we pre-registered Phase D accordingly (§5.1): margin $\delta_0 = 0.05$ as the reader-study standard, with verdicts reported honestly as inconclusive whenever the CI cannot exclude it.

As a preview, the power check ran the first real paired E2 comparison, using the published CNN score as “today’s machine” ($n = 105$ decided rows with both predictors, 99/6): human AUC = 0.460 [0.278, 0.677], CNN 0.641 [0.362, 0.898], paired Δ AUC(CNN – human) = +0.180 [–0.029, +0.419]. The direction favors the machine; at six negatives it decides nothing (and the CNN column mixes two catalogs’ scores, a per-catalog calibration caveat carried into Phase D).

4 Machine systems and the frozen gate

4.1 The leak-aware pool and the one-shot gate

All Phase C training and selection draws on one pool (`parity/build_train_splits.py`): 69,228 rows (2,657 graded, 2,313 grade-D human rejects, 64,507 random galaxies) *after* a firewall that excluded 801 rows — 769 by bench name (including the entire arm-1 augmentation strata) and 32 by position within 2” of an arm-2 target — with an audit asserting zero remaining collisions.

³arXiv:2604.02726, withdrawn by the author at v6. Of its 440 “deflector redshifts,” 385 are photometric; its own v5 validation reports 35% blind deflector- z recovery; its per-object catalog was never properly released. Our extraction (`parity/data/external/euclid.q1.nisp.csv`, 473 rows, each row’s `source_ref` carrying the warning) is retained for reference only and is deliberately not consumed by the bench builder. A deep-research brief’s “178 NISP z -pairs” traced to this paper’s v1 and should be treated as unsupported.

Splits (`train/valse1/gate`) are stratified by source $\times$ grade, seed 2026. The gate — 259 rows: 139 graded (including 79 A/B lenses), 60 grade-D, 60 random — is the frozen evidence set, scored once per frozen system and never used for selection. Two contrasts recur below: **HARD** (published A/B lenses vs. grade-D human rejects — the human-reject wall that defeated LENSJUDGE v1–v5) and **vetting** (A/B vs. grade-D + random).

4.2 C1a — the frozen CNN, and a score-provenance trap

The incumbent machine is the search’s own residual-network ensemble. Its gate number came with a trap we record so nobody refalls into it: graded rows in our tables carry the *reproduction* ensemble score while grade-D rows carry only the *published* probability, and mixing the two provenances across classes fabricates discrimination — gate HARD AUC = 0.847 mismatched vs. **0.646** [0.557, 0.734] like-for-like (published score on both sides). The pool now carries `p_pub` and `p_repro` as separate columns. Verdict: the frozen CNN has real but modest rank signal on the human-judged HARD contrast — far above LENSJUDGE v1’s ~ 0.5 , well below deployment grade.

4.3 C1b — representation probe: the wall is real for engineered features

A trained probe over the program’s engineered image features (soft-target logistic regression, $C = 0.01$, isotonic calibration fit on `valse1`) scores gate HARD AUC = 0.425 [0.330, 0.517] — below chance, as was every point of its selection grid — against 0.655 on the easy A-vs-random contrast and 0.716 on A/B/C-vs-random. This confirms, with a properly trained and gated probe, the v1 RepresentationKit result: tier-1 engineered features carry essentially no signal against human-rejected look-alikes. The probe scored both bench arms blind (912 and 2,605 rows) as a floor predictor for Phase D.

4.4 C2 — the 27B student: human-label SFT succeeds where distillation failed

The strongest system is a Qwen3.5-27B vision–language student fine-tuned *directly on the human grade catalog* — no Claude distillation, the explicit v4 lesson (a distilled student cannot exceed a teacher that §4.2 shows is weak; the argument does not bind when the teacher is the human catalog itself). Corpus (`finetune/build_corpus_desi.py`, leakage self-check passing): 12,250 dihedral-augmented training rows, 189 validation, 1,174 `valse1` selection rows. Training transplants the proven v5 run-3 recipe verbatim (bf16 LoRA $r=16/\alpha=32$ with the vision tower unfrozen, two epochs, 4-GPU DDP on a Perlmutter A100-80 node; job 55804993, 6h33m, 11 checkpoints). Selection (job 55848953) graded `valse1` with seven plateau checkpoints through the pre-registered grade-token log-probability score, $p_{\text{lens}} = P(A) + P(B)$ at the grade token: a monotone plateau with `ckpt-750` best (vetting 0.852, HARD 0.659). The one-shot gate (job 55859082) **held with no collapse: HARD** 0.644 [0.550, 0.733], **vetting** 0.817 [0.757, 0.873], against a zero-shot control (same protocol, base weights) of 0.530 [0.430, 0.621] and 0.576 — a training delta of +0.11 HARD and +0.24 vetting. Direct human-label SFT moves the tier-1 needle that v4’s Claude distillation could not.

4.5 C3 — the matched-inputs Claude grader: a real lever, and an honest non-replication

The human-baseline report ends by noting an input asymmetry: the graders saw multi-band composites with viewer context and redshift metadata, while LENSJUDGE’s deployed grader saw three rendered views of a bare 101-px cutout. C3 removes the asymmetry (`imaging/grader_matched.py`;

rubric `prompts/rubric_matched.md` = the five published grading criteria verbatim, in a relative-triage framing): full and zoom composites, a per-band $g|r|z$ montage, a 401-px ($\sim 105''$) wide-context view, and a metadata block (Tractor morphology gloss, region, the CNN score, spectroscopic/photometric redshifts where published, aperture grz magnitudes) — while *dropping* the machine-only affordances humans lacked (the residual view and the aperture-color JSON). The grader is Claude Sonnet 5 on the Agent SDK (Anthropic, 2025), sharing the deployed grader’s transport, JSON contract, and trace format.

On the 149-row selection slice (29A/30B/30C/40D/20 random; both arms Sonnet 5), matched inputs beat the deployed direct grader decisively: HARD 0.623 [0.518, 0.725] vs. 0.436 [0.312, 0.553], paired $\Delta\text{AUC} = +0.151$ [+0.020, +0.280]; on the full mix 0.640 vs. 0.401, $\Delta\text{AUC} = +0.161$ [+0.040, +0.281]. This is **the first statistically significant tier-1 lever in the program** — v1–v5 tried rubrics, agency, reasoning budgets, and representations and found none — and it says the missing ingredient was the human graders’ information set. (Robustness: matched graded 149/149 rows; direct only 115/149. Cost $\sim \$6$; a live six-candidate smoke first, \$0.121.)

The gate said something harder. With the selection frozen — matched mode, Sonnet 5, and a `valse1`-fit operating point ($p_{\text{lens}} > 0.050$ at target FPR = 0.20, plus an isotonic map, persisted as `outputs/c3_matched_op_point.json`) — the one-shot gate run (258/259 rows, \$4.36) scored **HARD** 0.487 [0.395, 0.577]: the `valse1` HARD level (0.623) *did not replicate*. That is selection optimism on a 149-row slice, and the disjoint-draw rule caught it — exactly what the discipline is for. The broad contrast held (0.633 [0.566, 0.699], A/B/C-vs-D+random), the frozen operating point ran hot on gate grade-D (FPR = 0.35 vs. the 0.20 target), and raw grades stayed cold ($\kappa_{\text{qw}} = 0.025$; mean p_{lens} on true grade-A 0.09 — the score ranks, the labels don’t). Standing conclusion: C3 is a real lens-vs-random *vetter*, not a HARD-contrast detector; the matched-vs-direct *relative* gain stands (it is paired, same rows, CI excluding zero), while the absolute HARD level is chance. Two caveats carry into Phase D: the metadata block includes the CNN score (fair — the humans had it — but Phase D carries CNN-alone to isolate any LLM added value), and deployment must go through the calibrated operating point, never the raw grade (the same lesson as the residual A/B, companion report⁴).

4.6 The Phase C scoreboard

Table 2 freezes the evidence. The human-label student ties the frozen CNN on the human-reject wall and posts the strongest vetting number the program has produced at DESI resolution; the zero-shot control isolates training as the cause; engineered features and the matched Claude grader sit at or below chance on HARD. Phase D therefore carries four frozen predictors to the bench: the **human grade** (the comparator), the **CNN** published probability, the **student-27B ckpt-750** log-probability score, and the **representation probe** (floor); C3 is documented reference, not run at bench scale.

5 Phase D: the parity comparison

5.1 Pre-registered methods

Everything in this subsection was fixed — predictors, code (`parity/phase_d_analysis.py`), seed, margin — before the bench parquets were scored; the analysis is a single scored run.

⁴An Honest Residual Image for Strong-Lens Vetting, <https://usfcs-edu.github.io/agentic-lensing/current/lensjudge-residual/>

system (frozen; gate scored once)	HARD: A/B vs. D	vetting
student-27B <code>ckpt-750</code> (grade-token logprob)	0.644 [0.550, 0.733]	0.817 [0.757, 0.873] ^a
frozen CNN (published p , like-for-like)	0.646 [0.557, 0.734]	—
zero-shot 27B (same protocol)	0.530 [0.430, 0.621]	0.576 ^a
C3 matched Sonnet 5 (frozen op point)	0.487 [0.395, 0.577]	0.633 [0.566, 0.699] ^b
representation probe (C1b)	0.425 [0.330, 0.517]	0.716 ^b

Table 2: Phase C final gate scoreboard (259-row frozen gate; 2,000-sample bootstrap, seed 2026). HARD = published A/B lenses vs. grade-D human rejects. Vetting: ^aA/B vs. grade-D+random; ^bA/B/C vs. grade-D+random (the two LLM-graded systems score all grades). Sources: §4.2–4.5; `parity/FINDINGS.md`.

predictor	score	n	AUC (truth) [95% CI]
human grade	ordinal A/B/C (comparator)	130	0.577 [0.400, 0.755]
frozen CNN	published probability	105	0.642 [0.360, 0.891]
student-27B	<code>ckpt-750</code> grade-token logprob	130	0.685 [0.538, 0.819]
rep. probe	C1b calibrated score (floor)	130	0.660 [0.442, 0.871]

Table 3: E2 absolute discrimination of confirmed-vs-refuted truth on the 130 decided arm-1 candidates (122 lens / 8 refuted); stratified bootstrap CIs (2,000, seed 2026). Values computed from `outputs/parity_phase_d.json (e2.*)`.

E2, primary. On the 130 decided arm-1 rows (122 lens, 8 refuted), each machine score is compared to the human ordinal grade ($A = 3$, $B = 2$, $C = 1$) as a paired predictor of truth. For each machine: $\Delta\text{AUC} = \text{AUC}_{\text{machine}} - \text{AUC}_{\text{human}}$ with a paired, class-stratified bootstrap (2,000 resamples, seed 2026) percentile CI and a two-sided DeLong p . Non-inferiority margin $\delta_0 = 0.05$ (the reader-study standard): *non-inferior* if the CI lower bound exceeds $-\delta_0$, *inferior* if the upper bound is below it, *inconclusive* otherwise. §3.3 says in advance that at 8 negatives most honest outcomes are estimation plus “inconclusive” at this margin; we report estimation regardless, and the verdict column is read jointly with the 0.24 noise floor.

E1, secondary. κ_{qw} between the student’s emitted letter grade and the published consensus on the arm-1 graded rows, read against the measured anchors (0.776 team-member upper bound / 0.42 mutual / 0.29 independent team).

Arm 2, context. Truth-referenced AUC on the 44 SLDE-B-decided rows (38/6) for the machine predictors; the Euclid panel’s `expert_score` scored on the same rows as a *resolution-privileged upper anchor* — it saw 0.1” pixels, so it is context, not a paired comparator; and grade-stratified student-score distributions across all 2,605 Euclid-field rows, the at-scale behavioral check.

5.2 Results

Status: the student bench `parquets (preds_bench_arm{1,2}.parquet)` are computing on `Perlmutter`; every value below is a placeholder macro filled from `outputs/parity_phase_d.json`.

Table 3 gives the absolute truth-referenced AUCs on the decided arm-1 rows; the human row should reproduce the Phase B estimate (0.577 under the power check’s bootstrap) up to resampling. Table 4 is the primary endpoint: each machine against the human grade, paired.

machine vs. human	ΔAUC [95% CI]	DeLong p	verdict at $\delta_0 = 0.05$
frozen CNN	+0.182 [−0.011, +0.425]	0.13	non-inferior
student-27B	+0.108 [−0.028, +0.241]	0.12	non-inferior
rep. probe	+0.084 [−0.193, +0.356]	0.59	inconclusive

Table 4: E2 primary endpoint: paired ΔAUC (machine − human) on the decided arm-1 rows, paired stratified bootstrap CI and two-sided DeLong p ; non-inferiority verdicts per §5.1, to be read against the ~ 0.24 decidable-margin floor of §3.3. Values computed from `outputs/parity_phase_d.json` (e2.d.*).

predictor (arm 2, $n = 44$ decided: 38 lens / 6 refuted)	AUC (truth) [95% CI]
student-27B <code>ckpt-750</code>	0.509 [0.268, 0.750]
rep. probe	0.693 [0.438, 0.912]
Euclid panel <code>expert_score</code> (0.1'' anchor)	0.833 [0.600, 1.000]

Table 5: Arm-2 context: truth-referenced AUC on the SLDE-B-decided rows (Euclid Collaboration et al., 2025a). The Euclid panel score is a resolution-privileged upper anchor, not a paired comparator: it was assigned from 0.1'' VIS pixels, the machines see DR10 ground-based pixels of the same sky. Values computed from `outputs/parity_phase_d.json` (arm2.*).

On E1, the student’s emitted letter grades score $\kappa_{\text{qw}} = 0.044$ against the published consensus ($n = 162$ graded rows) — to be read against the 0.29/0.42/0.776 human ladder; the Phase C gate ($\kappa_{\text{qw}} = 0.025$ for C3) and every prior LENSJUDGE generation predict this lands far below the independent-team anchor, with the rank score, not the label, carrying whatever signal exists.

Arm 2 (Table 5) contextualizes the same predictors on the Euclid field at DESI resolution, where the truth tier is modeling-based and the refutations are SLDE-B’s six. At scale, the student’s mean score by Euclid panel grade is 0.049 (A), 0.035 (B), 0.023 (C) across the 2,605 rows — the monotonicity (or not) of that triple is the at-scale sanity check on whether the score tracks the panel’s confidence ordering.

6 Discussion

What “parity” can mean when the comparator is at chance. The most consequential number in this program may be Table 1’s last row: on the pool where truth exists, the human grade discriminates confirmed from refuted at an AUC statistically consistent with 0.5. E2 non-inferiority to that comparator is therefore a *low* bar — a machine could clear it while being useless — and a machine that clears it has *not* thereby shown it “grades like a trained human.” What the humans demonstrably did well is upstream of the grade: selecting, from millions of galaxies, a candidate list whose decided follow-ups confirm at $\gtrsim 90\%$ in every grade. The honest reading of any Phase D non-inferiority verdict is symmetric and modest: on curated follow-up targets, the machine’s score carries at least as much (little) incremental truth signal as the expert fine grade. The agreement endpoint E1, with its measured 0.29/0.42/0.776 ladder, remains the harder and more human-flavored target — and every machine system to date fails it badly.

The refuted-negative bottleneck. Fourteen per-object refutations exist, anywhere, for this task — eight spectroscopic in the program’s own follow-up, six by Euclid SLDE-B modeling. That single number sets the 0.24 margin floor, makes every CI in §5.2 wide, and explains why the two Euclid integrity findings (§3.2) matter beyond bookkeeping: apparent truth that dissolves under audit (a

withdrawn paper, unpublished verdicts) would have manufactured false statistical power. The path to a real verdict is already scheduled science: the grade-agnostic DESI Secondary Target Program (2,157 systems, selection-clean by construction) and Euclid-Collaboration per-object adjudications at DR1 scale would bring the decidable margin to 0.13–0.078 — reader-study territory. The bench, firewall, and pre-registered analysis here are built to be re-run unchanged on that data.

Honest scope. Everything above is conditional on a curated pool: follow-up campaigns chose their best targets, so the decided set inherits each campaign’s selection function and compresses the score range on which both human and machine are compared; purities are upper bounds. The comparison is DESI-resolution only — the companion Part I result (grade C largely re-grades to A/B at 0.1”) says resolution, not judgment, dominates at the margins. The human predictor is a three-level ordinal, structurally disadvantaged against continuous scores in AUC (a granularity, not a competence, artifact — one more reason E2 verdicts should not be over-read). Arm-2 truth is modeling-plus-expert, not spectroscopic, and its refutations are SLDE-B grade-B systems — a different, easier negative class than arm 1’s spectroscopically dissolved mimics. And the student’s training pool, however firewalled, is drawn from the same survey and grading program as the bench: this is parity *within* the program’s distribution, not a transfer claim.

What to deploy. The program’s recommendation is unchanged by whichever way the pending verdicts land, because it follows from the structure of the results. Machine scores are now good enough to *vet at scale* — the student-27B’s 0.817 vetting AUC through a calibrated, FPR-matched operating point (never a raw grade: C3’s hot operating point and cold labels, and the residual A/B before it, are two independent demonstrations) — so the defensible pipeline is CNN → student score for triage and ranking, with expert graders spending their attention on the survivors. The A/B/C letter grades stay human: no system approaches the E1 ladder, and the grade’s published meaning — two named experts’ consensus — is not something a score distribution can impersonate. “Human parity,” if the phrase is used at all, should be used the way this report uses it: as a pre-registered, truth-referenced, margin-explicit statement about a specific pool, with the selection-vs-gate discipline that caught our own best lever inflating itself.

7 Reproducibility

All inputs are published catalogs, the program’s own graded tables, and the committed artifacts below; every random draw is seed 2026 and the bench CSVs are SHA-pinned. The running factual log is `parity/FINDINGS.md`.

Phase A (companion report²) `parity/fetch_vizier_paper2.py`; `parity/intergrader_stats.py`; `parity/grade_purity_table.py`; `parity/human_baseline_summary.py`; truth tables in `parity/data/external/*.csv`, with rerunnable Euclid builders in `parity/external_builders/` (`build_euclid_q1_modeling.py`, `build_euclid_q1_nisp.py`).

Phase B `eval/build_parity_bench.py` → `outputs/parity_bench_arm{1,2}.csv` (912 / 2,605 rows); `parity/stage_parity_cutouts.py` (cutout staging, complete on both arms); `parity/power_check.py` → `outputs/parity_power_check.csv`.

Phase C `parity/build_train_splits.py` → `outputs/parity_train_pool.csv` (with the firewall audit); `parity/train_rep_probe.py` and `parity/score_bench_rep.py` → `outputs/rep_probe/`; the matched grader `imaging/grader_matched.py` with `prompts/rubric_`

matched.md, parity/smoke_matched.py, parity/build_c3_valsel_slice.py, parity/c3_select_analysis.py, and the frozen operating point outputs/c3_matched_op_point.json; the student pipeline finetune/build_corpus_desi.py (corpus, label, and gate subcommands), finetune/train_lora.sh, and finetune/ljdesi_train27b.slurm, with the full runbook in finetune/PERLMUTTER_DESI_RUN.md (Perlmutter jobs 55804993 train, 55848953 selection, 55859082 gate; artifacts under \$SCRATCH/ljdesi/: ckpt_desi_27b/v0-*/checkpoint-750 and its merge, preds_valsel*.parquet, preds_gate_student750.parquet, preds_gate_zeroshot.parquet); finetune/wise_ft.py (weight interpolation, documented but not deployed here).

Phase D parity/phase_d_analysis.py → outputs/parity_phase_d.json, which fills every results macro in this document once outputs/preds_bench_arm{1,2}.parquet land from Perlmutter.

References

- Shrihan Agarwal, Xiaosheng Huang, et al. DESI Strong Lens Foundry III: Keck Spectroscopy for Strong Lenses Discovered Using Residual Neural Networks. *arXiv e-prints*, 2025.
- Anthropic. Claude Agent SDK for Python. <https://github.com/anthropics/claude-agent-sdk-python>, 2025. v0.2.x; rides on the Claude Code CLI for transport and auth.
- Tania M. Barone, Keerthi Vasan G. C., et al. The AGEL Survey Data Release 2: A Gravitational Lens Sample for Galaxy Evolution and Cosmology. *arXiv e-prints*, 2025.
- R. Cañameras, S. Schuldt, S. H. Suyu, et al. HOLISMOKES VI: New galaxy-scale strong lens candidates from HSC. *Astronomy & Astrophysics*, 653:L6, 2021. doi: 10.1051/0004-6361/202141758.
- A. Dey et al. Overview of the DESI Legacy Imaging Surveys, 2019.
- Euclid Collaboration. Euclid Quick Data Release (Q1): The Strong Lensing Discovery Engine (data). Zenodo, 2025.
- Euclid Collaboration, K. Rojas, T. E. Collett, J. A. Acevedo Barroso, et al. Euclid Quick Data Release (Q1): The Strong Lensing Discovery Engine B – Early strong lens candidates from visual inspection of high velocity dispersion galaxies. *arXiv e-prints*, 2025a.
- Euclid Collaboration, M. Walmsley, P. Holloway, et al. Euclid Quick Data Release (Q1): The Strong Lensing Discovery Engine A – System overview. *arXiv e-prints*, 2025b.
- X. Huang, C. Storfer, V. Ravi, et al. Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey. *The Astrophysical Journal*, 894(1):78, 2020. doi: 10.3847/1538-4357/ab7ffb.
- X. Huang, C. Storfer, A. Gu, et al. Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys. *The Astrophysical Journal*, 909(1):27, 2021. doi: 10.3847/1538-4357/abd62b.
- X. Huang, S. Baltasar, et al. DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens. *ApJ*, 998, 2026.
- X. Huang et al. DESI Strong Lens Foundry II: DESI Spectroscopy. *arXiv e-prints*, 2025.

- J. Inchausti Reyes, C. Storfer, X. Huang, et al. A Search for Strong Gravitational Lenses in DESI-DR10 with Two Neural-Network Architectures. *arXiv e-prints*, 2025.
- Emerald Lin, Ivonne Toro Bertolla, et al. DESI Strong Lens Foundry IV: Spectroscopic Confirmation of DESI Lens Candidates with VLT/MUSE. *arXiv e-prints*, 2025.
- K. Rojas, T. E. Collett, D. Ballard, et al. The impact of human expert visual inspection on the discovery of strong gravitational lenses. *Monthly Notices of the Royal Astronomical Society*, 523(3):4413–4430, 2023. doi: 10.1093/mnras/stad1680.
- Yiping Shu, Shen Li, et al. Confirming HSC strong lens candidates with DESI Spectroscopy. I. Project overview and first results. *arXiv e-prints*, 2025.
- A. Sonnenfeld, J. H. H. Chan, Y. Shu, et al. Survey of Gravitationally-lensed Objects in HSC Imaging (SuGOHI). I. *Publications of the Astronomical Society of Japan*, 70:S29, 2018. doi: 10.1093/pasj/psx062.
- C. Storfer, X. Huang, et al. New Strong Gravitational Lenses from the DESI Legacy Imaging Surveys DR9. *The Astrophysical Journal Supplement Series*, 274(2):16, 2024. doi: 10.3847/1538-4365/ad646e.
- Kim-Vy H. Tran, Anishya Harshan, et al. The AGEL Survey: Spectroscopic Confirmation of Strong Gravitational Lenses in the DES and DECaLS Fields Selected Using Convolutional Neural Networks. *AJ*, 164:148, 2022.